SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1507

COURSE OUTCOME COVERED
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct
I __Narmatha J (192521341)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: _NARMATHA J_

Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

ANSWERS:

Question 1: Media Company – Video Streaming Platform

Scenario:

A media company wants to migrate its video streaming platform to the cloud. The platform should support millions of concurrent users with minimal buffering and uninterrupted video streaming.

a) Design a Cloud Architecture using Scalability and Caching:

To meet the high demand, the company should deploy the application using a multi-tier cloud architecture. Video files should be stored in cloud object storage, while application servers should run behind a Load Balancer. A Content Delivery Network (CDN) should cache videos at edge locations so users receive content from the nearest server. Frequently accessed data can also be stored in Redis or Memcached to reduce database access. Auto Scaling Groups should automatically add or remove servers based on traffic.

b) Explain how Load Balancing and Auto-Scaling ensure smooth streaming:

The Load Balancer distributes user requests evenly among multiple application servers, preventing any single server from becoming overloaded. During peak hours, Auto Scaling automatically launches additional servers to handle increased traffic. When demand decreases, unnecessary servers are removed. This maintains high performance, minimizes buffering, and ensures uninterrupted video streaming.

c) Cost Optimization Strategies:

- Use Auto Scaling so resources are used only when needed.
- Store frequently accessed videos in CDN cache to reduce bandwidth costs.
- Move old or rarely viewed videos to low-cost archive storage.
- Use Reserved Instances or Savings Plans for predictable workloads.
- Continuously monitor cloud resources and remove unused services to reduce cost.

Question 2: Fintech Startup – Multi-Cloud Container Deployment

Scenario:

A fintech startup deploys applications using containers across multiple cloud providers. The company needs consistent deployments, high availability, and efficient application management.

a) Propose a solution using Container Orchestration and Multi-Cloud:

The startup should package applications using Docker containers and manage them using Kubernetes. Kubernetes clusters can be deployed across multiple cloud providers such as AWS, Azure, and Google Cloud. Terraform or Infrastructure as Code (IaC) can be used to deploy identical infrastructure across all clouds. This ensures consistency, simplifies deployment, and provides high availability.

b) Explain how Service Discovery and Scaling are handled:

Kubernetes provides Service Discovery through internal DNS, allowing containers to communicate without manually configuring IP addresses. Horizontal Pod Autoscaler automatically increases or decreases the number of running containers based on CPU or memory usage. Load Balancers distribute requests among available containers to maintain application performance.

c) Benefits and Risks of Multi-Cloud Deployment:

Benefits:

- High availability if one cloud provider fails.
- Prevents vendor lock-in.
- Better disaster recovery.
- Improved flexibility and reliability.
- Ability to choose the best services from different providers.

Risks:

- More complex management.
- Increased operational costs.
- Data synchronization challenges.
- Security policy differences.
- Possible network latency between cloud providers.

Question 3: E-Learning Platform – Backup and Disaster Recovery

Scenario:

An e-learning platform loses important data due to accidental deletion. The organization requires a reliable backup and disaster recovery solution to restore services quickly.

a) Design a Solution using Backup, Replication and Versioning:

The platform should enable automatic cloud backups and schedule regular database snapshots. Object Versioning should be enabled so deleted files can be restored easily. Data should also be replicated across multiple cloud regions to ensure a backup is always available even if one region fails. Backup copies should be stored separately for additional protection.

b) Explain how RTO and RPO are achieved:

- **Recovery Time Objective (RTO)** is achieved by maintaining standby systems and automated failover, allowing services to recover quickly after a failure.
- **Recovery Point Objective (RPO)** is achieved through continuous replication and frequent backups, ensuring only a minimal amount of data is lost.

c) Steps to Ensure Business Continuity:

- Enable automatic cloud backups.
- Replicate data across multiple regions.
- Configure disaster recovery infrastructure.
- Regularly test backup restoration.
- Monitor backup status continuously.
- Maintain a documented disaster recovery plan and train administrators.

Question 4: Government Agency – Hybrid Cloud

Scenario:

A government agency adopts a hybrid cloud model where sensitive applications remain on-premises due to compliance requirements, while non-sensitive services are deployed in the public cloud.

a) Design a Hybrid Cloud Architecture:

Sensitive databases and critical applications should remain in the on-premises data center. Public-facing applications such as websites and citizen services can run in the public cloud. Both environments should be connected using a secure VPN or dedicated private connection. APIs should securely exchange data between the on-premises and cloud environments while maintaining compliance.

b) Explain Identity Management and Encryption:

A centralized Identity and Access Management (IAM) system should authenticate users across both environments. Multi-Factor Authentication (MFA) should be enabled for all administrators. Data should be encrypted at rest using AES encryption and during transmission using SSL/TLS. Role-Based Access Control (RBAC) should ensure users access only authorized resources.

c) Challenges in Hybrid Cloud:

- Managing infrastructure across multiple environments.
- Maintaining consistent security policies.
- Compliance with government regulations.
- Network latency between cloud and on-premises systems.
- Data synchronization and integration challenges.
- Increased administrative complexity.

Question 5: SaaS Provider – High Availability

Scenario:

A SaaS provider requires its application to remain available even if an entire cloud region becomes unavailable. The organization must provide continuous service without affecting customers.

a) Propose a Multi-Region Deployment Strategy:

The application should be deployed across multiple cloud regions with identical infrastructure in each region. Global Load Balancers should distribute user requests to the nearest healthy region. Databases should be deployed with replication across regions, and Auto Scaling should automatically increase resources based on demand. This architecture ensures continuous availability even during regional failures.

b) Explain Data Replication and Failover:

Critical application data should be replicated continuously between multiple regions. Health monitoring services should detect failures automatically. If the primary region becomes unavailable, automatic failover should redirect users to the secondary region with minimal downtime. Real-time replication ensures that users experience little or no data loss.

c) Monitoring and Alerting Strategies:

- Continuously monitor CPU, memory, storage, and network utilization.
- Configure alerts for server failures, high resource usage, and application errors.
- Monitor application response time and database performance.
- Use centralized logging for troubleshooting.
- Conduct regular health checks and disaster recovery drills to ensure system readiness.